FIELD / 2458

A small laboratory, built as an entity-component-system

Tair Asset | AIB-2401 | Student ID 242458 | PA4

Six primitive-built exhibits occupy a 4 m ring. ID digits 8, 5, 4, 2 select science lab, pulse, 1.5-second gaze fuse and ring layout. Core A-Frame 1.8.0 supplies Three.js r184; three original CC0 textures preload through a-assets. The three custom components and gallery system use the x2458- prefix.

E1 / A component through time

Prediction. init runs once; the first update has no previous configuration. Changing amplitude calls update again. Pause suppresses motion; play resumes it. Removing the component calls remove; subsequently deleting its entity cannot remove that same component twice.

Timestamp, ms	Observed method	Previous configuration
8951.8	init	-
8952.4	update	{}
8953.2	play	-
8954.5	tick	-
9558.7	update	{"type":"pulse","amplitude":0.12,"period":4,"phase":0}
9562.8	pause	-
9565.1	play	-
9992.3	tick	-
10074.5	pause	-
10076.2	remove	-

Result and explanation. The log follows the prediction. oldData is empty on initialization and contains the previous parsed values after setAttribute. Motion keeps base transforms and an accumulated phase; period changes affect future phase advance. Amplitude/type changes preserve the current pose through a decaying offset. Local time stops during pause. Tick logging is sampled to avoid flooding the console.

E2 / Rechecking the chapter

Prediction. Reusing the chapter loader will recreate runtime objects on a configuration change. Calling the asset manager a system does not make it an AFRAME.registerSystem instance.

Result. A test fixture reconstructed from supplied Ch. 8 pp. 12-15 replaced both texture and material after changing src; the previous texture was not disposed. Before and after, this.data contained Texture, MeshLambertMaterial and Mesh objects. In the corrected loader, this.data contains configuration only; objects live on the instance. The actual system list contains x2458-gallery but no a-assets.

Explanation. Parsed, serializable attributes describe configuration; mesh/material/texture ownership needs explicit lifecycle cleanup. A-assets is an AAssets custom element extending ANode, with asset-loading behaviour. It is an asset-management subsystem, not the scene singleton created by registerSystem.

E3 / A floor at a grazing angle

Prediction. Nearest minification will alias the repeated grid. Mipmaps blend distant detail; anisotropy should preserve oblique detail better.

Setting	Minification	Magnification	Anisotropy
Chapter	NearestFilter	LinearFilter remains	16
Three.js defaults	LinearMipmapLinearFilter	LinearFilter	1
Chosen	LinearMipmapLinearFilter	LinearFilter	8

Result. The same fixed camera and paused scene produced the three attached images. Nearest gives harder, less stable distant grid detail; default mipmaps soften it; the chosen setting balances smoother distance transitions with oblique detail. Static images show spatial aliasing, not temporal shimmer by themselves.

Explanation. Minification handles a texel footprint smaller than a screen pixel; magnification enlarges texels. Mipmaps provide prefiltered smaller images. The chapter writes maxFilter, which is absent from a fresh Three.js Texture. The valid property is magFilter; assigning maxFilter merely adds an unused JavaScript field. The corrected loader uses magFilter and clamps anisotropy to the device limit. Ground repetition is 16 x 16.

E4 / What the scene builds

Prediction. A-scene will create Three.js rendering objects and manage session setup that PA3 handled explicitly.

Console inspection	Observed result
sceneEl.renderer.isWebGLRenderer	True
sceneEl.object3D.type	Scene
sceneEl.camera.type	PerspectiveCamera
sceneEl.renderer.xr	Present

Explanation. In A-Frame v1.8.0 src/core/scene/a-scene.js, enterVR starts at line 295; lines 320-329 request the session and pass it to renderer.xr.setSession. PA3 explicitly requested immersive-ar, negotiated features, obtained reference spaces and cleaned up hit testing. A-Frame wraps common scene/session work. A company building a specialised CAD renderer may prefer raw Three.js; a small interactive exhibition benefits from A-Frame composition and reusable components.

Measurement context

The desktop experiments use the actual browser and Intel UHD graphics reported by WebGL. Tests are automated, not physical-headset measurements. The VR check uses official Immersive Web Emulator 2.0 with a Meta Quest 3 profile; renderer.xr.isPresenting and a gaze selection were both observed. No physical headset comfort or 72 Hz claim is made.

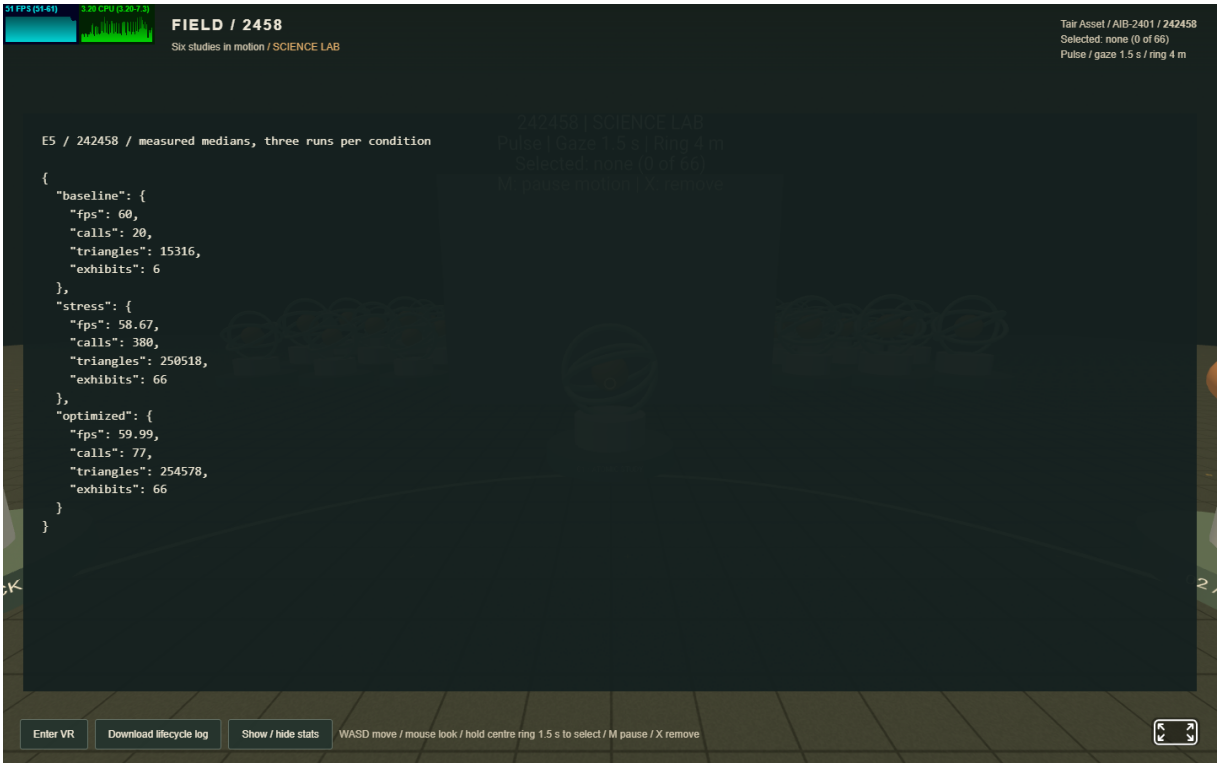
E5 / Rendering more exhibits

Prediction. Sixty added exhibit entities will increase draw calls and triangle work; instancing compatible meshes should reduce submissions without removing exhibits.

Scene	Exhibits	FPS	Draw calls	Triangles
Original	6	60	20	15316
+60 exhibits	66	58.67	380	250518
Instanced	66	59.99	77	254578

Result. Values are medians of three three-second desktop runs per condition, after four-second warm-ups, with stats enabled. Compatible geometry/material groups become InstancedMesh objects, while the exhibit entities remain registered. Additional instances can render outside the original per-object frustum culling, so triangles need not fall with draw calls.

Explanation. Draw calls incur CPU/driver submission and state-change costs. VR must supply two views within a tighter frame budget, making unnecessary submissions costly. Instancing reduces calls rather than pretending to reduce model complexity. FPS is a sampled observation, not a guarantee for every machine.



E5: actual browser counters displayed by the experiment harness; ID remains visible.

E6 / Comfort and design

Prediction. An open layout, readable labels and deliberate gaze selection should reduce accidental interaction. **Result.** Exhibit centres are 4 m apart; the camera starts at 1.6 m. Measured status-line height at the initial viewing distance gives approximately 2.21 degrees. The fuse is 1.5 seconds. The emulator check confirms selection and removal of WASD in VR. **Explanation.** Angular size is 2 atan(line height / (2 distance)). Labels face the centre; the visible ring, selected text and modest pulse amplitude provide feedback. Smooth WASD movement can cause discomfort, so it is disabled in VR and restored on exit. Visitors look around from a stationary position. No physical-headset comfort study was performed.

Development log

The gaze ring was present but world geometry could paint over it.
A screenshot and mesh inspection exposed the render-order issue.
Commit ade0824 draws the ring last and freezes resumed timing.
Child-event regression checks found double selection; dad2353 stops propagation.
fac6055 snapshots oldData before the framework reuses it.

E4 / Browser inspection



Actual renderer, scene and camera observations from the running application.

Observed lifecycle and chapter data



E1: timestamped hook calls for the required sequence. The final entity deletion follows removal of its motion component.



E2: before/after runtime inspection of the chapter fixture, rendered by the test harness.

Three settings, one camera



E3 / Chapter: NearestFilter, anisotropy 16; maxFilter does not change magnification.

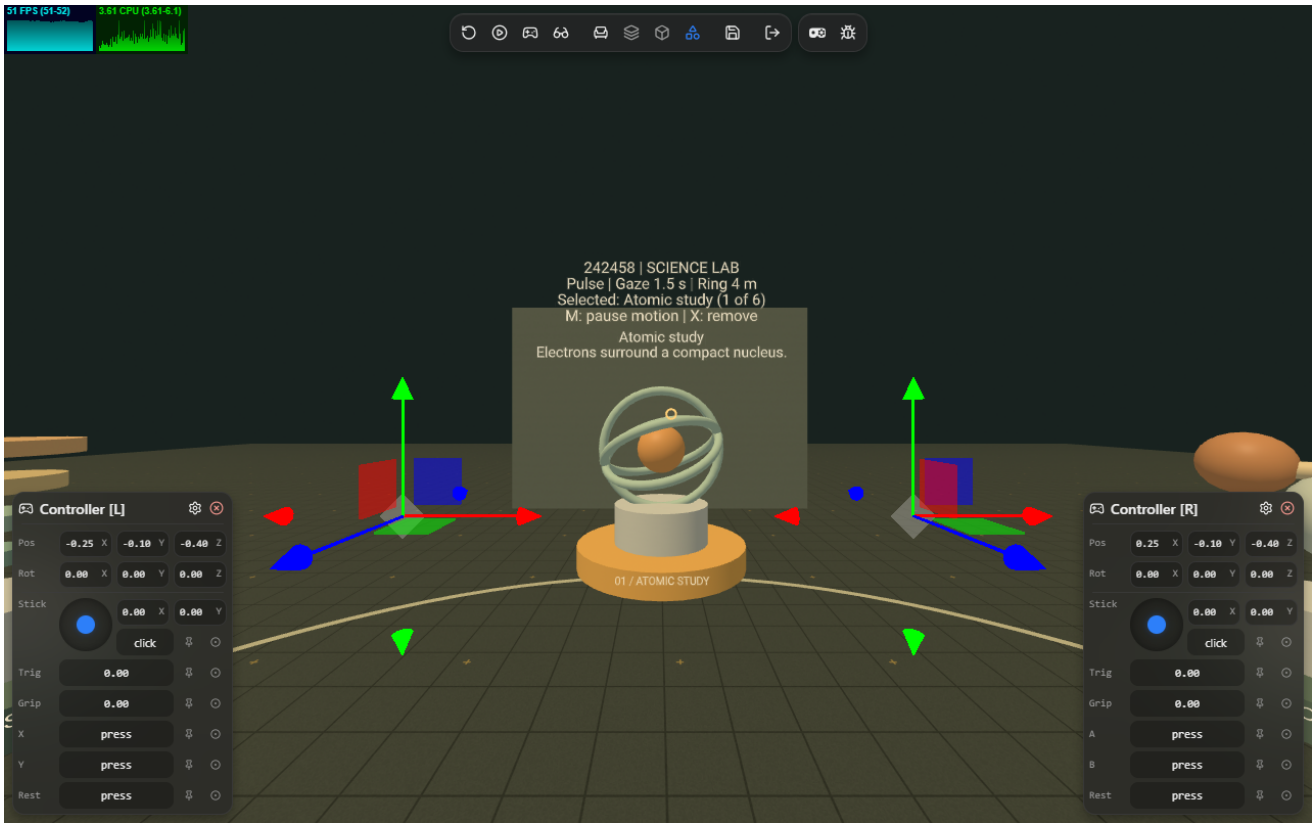


E3 / Three.js defaults: trilinear mipmap minification, linear magnification, anisotropy 1.

Chosen filtering and VR selection



E3 / Chosen: trilinear mipmaps, linear magnification, supported anisotropy up to 8.



VR / Official emulator, active WebXR session, gaze-selected Atomic study. This is emulation, not a physical headset recording.

Sources: supplied PA4 assignment; supplied Baruah Ch. 8 PDF pp. 6, 12-15, 18 and corresponding presentation; A-Frame v1.8.0 a-assets.js and a-scene.js inspected for E2/E4. Raw measurements and captures are retained locally.